

$$17) 12\% \text{ of } 25 = 3$$

$$18) 7\% \text{ of } 300 = ?$$

$$7\% \text{ of } 200 = 21$$

$$19) 32\% \text{ of } 250 = ?$$

$$32\% \text{ of } 250 = 80$$

20

$$20) 65\% \text{ of } 200 = ?$$

$$65\% \text{ of } 200 = 130$$

$$10\% = 20$$

$$50\% = 100$$

$$1\% = 2$$

$$5\% = 10$$

DAY-16

PERCENTAGE BASIC QUESTION

1) IF 50% of $P = 25\%$ of θ , then $P = x\%$ of θ . Find x .

$$2P = \theta$$

$$P = \frac{\theta}{2}$$

$$P = x\% \text{ of } Q$$

$$\frac{Q}{P} = \frac{x}{100} \times \frac{100}{50}$$

$$\boxed{x = 50}$$

2) 20% of $(P+Q) = 50\%$ of $(P-Q)$, then

Find $P:Q$

$$20\% \text{ of } (P+Q) = 50\% \text{ of } (P-Q)$$

$$2(P+Q) = 5(P-Q)$$

$$2P+2Q = 5P-5Q$$

$$2Q+5Q = 5P-2P$$

$$7Q = 3P$$

$$\boxed{\frac{P}{Q} = \frac{7}{3}}$$

3) If 90% of $A = 30\%$ of B and $B = 2x\%$ of A , then the value of x is

$$90\% \text{ of } A = 30\% \text{ of } B$$

$$3A = B$$

$$A = \frac{B}{3}$$

$$B = \frac{2x}{100} \times \frac{B}{3}$$

$$B \times \frac{300}{B} = 2x$$

$$\boxed{x = 150}$$

4) IF 40% of $(A+B) = 60\%$ of $(A-B)$

$$\text{then } \frac{(2A - 3B)}{(A+B)}$$

$$\frac{2}{40\%} \text{ of } (A+B) = \frac{3}{60\%} \text{ of } (A-B)$$

$$2(A+B) = 3(A-B)$$

$$2A + 2B = 3A - 3B$$

$$2B + 3B = 3A - 2A$$

$$5B = A$$

$$B = \frac{A}{5}$$

$$\frac{(2A - 3B)}{A+B} = \frac{2(5B) - 3B}{5B+B} = \frac{10B - 3B}{6B}$$

$$\frac{7B}{6B} = \frac{7}{6}$$

5) 20% of $a = 80\%$ of b then $(b+a)/b$

$$20\% \text{ of } a = \frac{4}{20\%} \text{ of } b$$

$$a = 4b$$

$$\frac{b+a}{b-a} \Rightarrow \frac{b+4b}{b-4b} \Rightarrow -\frac{5b}{3b} \Rightarrow -\frac{5}{3}$$

6) 20% of $(A+B) = 50\%$ of B , then

$$20\% \text{ of } (A+B) = 50\% \text{ of } B \quad \left(\frac{2A+B}{2A+B} \right) \text{ is}$$

$$2(A+B) = 5B$$

$$2A + 2B = 5B$$

$$2A = 3B$$

$$A = \frac{3B}{2}$$

$$\left(\frac{2A+B}{2A+B} \right) \Rightarrow \frac{3B-B}{3B+B}$$

$$\Rightarrow \frac{2B}{4B}$$

$$= \frac{1}{2}$$

7) If x is 20% less than y , then find

the value of $(y-x)/y$ and $x/(x-y)$

Let
 $\frac{x}{80}, \frac{y}{100}$

$$\left(\frac{y-x}{y} \right) \Rightarrow \frac{100-80}{100}$$

$$= \frac{20}{100}$$

$$= \frac{1}{5}$$

$$\left(\frac{x}{x-y} \right) \Rightarrow \frac{80}{80-100}$$

$$\Rightarrow \frac{80}{-20} = -4$$

8) IF 8% of $x = 4\%$ of y , then

20% of x is

$$\cancel{8}\% \text{ of } x = \cancel{4}\% \text{ of } y$$

$$\boxed{2x = y} \quad \boxed{x = \frac{y}{2}}$$

20% of x

$$\cancel{20}\% \text{ of } \frac{y}{2}$$

$$\boxed{10\% \text{ of } y}$$

9) IF 60% of $A = 30\%$ of B ,

$B = 40\%$ of C , $C = x\%$ of A ,

then the value of x

$$\cancel{60}\% \text{ of } A = \cancel{30}\% \text{ of } B$$

$$2A = B$$

$$x\% A = \frac{40\%}{\cancel{100}\% \times 5} \times C$$

$$A = \frac{C}{5}$$

$$C = 2\% \text{ of } A$$

$$C = \frac{x}{100} \times \frac{C}{5}$$

$$x = 500$$

10)